



AQ6.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Suppose $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x, 0)$. Which of the following options are correct?

☐

The matrix corresponding to T with respect to the standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1, 0), (0, 1)\}$, for both the domain and co-domain is $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

☐

The matrix corresponding to T with respect to the standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1, 0), (0, 1)\}$, for both the domain and co-domain is $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

☐

T is neither one-one nor onto.

☐

T is one-one but not onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The matrix corresponding to T with respect to the standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1, 0), (0, 1)\}$, for both the domain and co-domain is $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

T is neither one-one nor onto.

1 point

If the matrix corresponding to a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, with respect to standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1, 0), (0, 1)\}$, for both the domain and co-domain, is $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$. Which of the following is the appropriate definition of T ?

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$T(x, y) = (2x + y, 3x + 4y)$

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$T(x, y) = (2x + y, 3x - 4y)$

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$T(x, y) = (x + 4y, 2x + 3y)$

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$T(x, y) = (2x + 3y, x + 4y)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(x, y) = (2x + 3y, x + 4y)$

Let $W = \{(x, y, z) \mid x = 2y + z\}$ be a subspace of $\mathbb{R}^3$. Let

$\beta = \{(2, 1, 0), (1, 0, 1)\}$ be a basis of W . Let $T: W \rightarrow \mathbb{R}^2$ be a linear transformation such that $T(2, 1, 0) = (1, 0)$ and $T(1, 0, 1) = (0, 1)$. Answer the questions 3, 4 and 5 using the given information.

1 point



Which of the following is the appropriate definition of T ?

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$$T(x, y, z) = (x, y) \quad T(x, y, z) = (x, y)$$

☐

$$T(x, y, z) = (x, z) \quad T(x, y, z) = (x, z)$$

☐

$$T(x, y, z) = (y, z) \quad T(x, y, z) = (y, z)$$

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$$T(x, y, z) = (x - y - z, x - 2y) \quad T(x, y, z) = (x - y - z, x - 2y)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$T(x, y, z) = (y, z) \quad T(x, y, z) = (y, z)$$

$$T(x, y, z) = (x - y - z, x - 2y) \quad T(x, y, z) = (x - y - z, x - 2y)$$

1 point

Choose the correct options.

☐

T is one to one but not onto.

☐

T is onto but not one to one.

☐

T is neither one to one nor onto.

☐

T is an isomorphism.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is an isomorphism.

1 point

What will be the matrix representation of T with respect to the basis β for W and $\gamma = \{(1, 1), (1, -1)\}$ for R^2 ?

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$$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \end{bmatrix}$$

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$$\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \end{bmatrix}$$

1 point

Choose the set of correct options.

☐

Let $u = (3, 1, 0)$, $v = (0, 1, 7)$, and $w = (3, 0, -7)$. There is a linear

transformation $T: R^3 \rightarrow R^3$ such that $T(u) = T(v) = (0, 0, 0)$ and that $T(w) = (5, 1, 0)$.

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Let $u = (2, 1, 0)$, $v = (1, 0, 1)$, and $w = (3, 5, 6)$. There is a linear transformation $T: R^3 \rightarrow R^2$ such that $T(u) = (1, 0)$, $T(v) = (0, 1)$ and that $T(w) = (5, 6)$.

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Let $u = (1, 0, 0)$, $v = (1, 0, 1)$, and $w = (0, 1, 0)$. There is a linear transformation $T: R^3 \rightarrow R^3$ such that $T(u) = (0, 0, 0)$, $T(v) = (0, 0, 0)$, $T(w) = (0, 0, 0)$, and $T(1, 4, 2) = (2, 4, 1)$.

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Let $u = (1, 0)$, and $v = (0, 1)$. There is a linear transformation $T: R^2 \rightarrow R^2$ such that $T(u) = (\pi, 1)$, and $T(v) = (1, e)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let $u = (2, 1, 0)$, $v = (1, 0, 1)$, and $w = (3, 5, 6)$. There is a linear transformation $T: R^3 \rightarrow R^2$ such that $T(u) = (1, 0)$, $T(v) = (0, 1)$ and that $T(w) = (5, 6)$.

Let $u = (1, 0)$, and $v = (0, 1)$. There is a linear transformation $T: R^2 \rightarrow R^2$ such that $T(u) = (\pi, 1)$, and $T(v) = (1, e)$.

Level 2

1 point

Choose the set of correct options.

☐

If the Identity matrix of order 2 is the matrix representation of a linear transformation $T: R^2 \rightarrow R^2$ with respect to the standard ordered basis for both the domain and co-domain, then it is also the matrix representation of T^T with respect to any other basis β for both the domain and co-domain.

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Let T be an isomorphism between two vector spaces, and A be the matrix representation of T with respect to some basis. Then the nullity of A is 0.

☐

If $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ is the matrix representation of $T: R^2 \rightarrow R^2$ with respect to the standard ordered basis of R^2 , for both the domain and co-domain, then the matrix representation of $T \circ T^T \circ T$ with respect to the same bases will also be A .

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If $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ is the matrix representation of $T: R^2 \rightarrow R^2$ with respect to the standard ordered basis of R^2 , for both the domain and co-domain, then the matrix representation of $T \circ T^T \circ T$ will be the identity matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If the Identity matrix of order 2 is the matrix representation of a linear transformation $T: R^2 \rightarrow R^2$ with respect to the standard ordered basis for both the domain and co-domain, then it is also the matrix representation of T^T with respect to any other basis β for both the domain and co-domain.

Let T be an isomorphism between two vector spaces, and A be the matrix representation of T

with respect to some basis. Then the nullity of AA is 0.

If $AA = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ is the matrix representation of $T: R^2 \rightarrow R^2$ with respect to the

standard ordered basis of R^2 , for both the domain and co-domain, then the matrix representation of $T \circ T \circ T$ with respect to the same bases will also be AA .

Let $\beta = \{(1, 0), (0, 1)\}$ and $\gamma = \{(1, 1), (1, -1)\}$ be two bases of R^2 . If $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is the matrix representation of a linear transformation $T: R^2 \rightarrow R^2$

with respect to β for both the domain and the co-domain. Answer questions 8 and 9 using the information given above.

1 point

What is the matrix representation of T with respect to γ for the domain and β for the co-domain?

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$$\begin{bmatrix} a + b & c + d \\ a - b & c - d \end{bmatrix} [a + ba - bc + dc - d]$$

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$$\begin{bmatrix} a - b \\ c - d \end{bmatrix} [ac - b - d]$$

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$$\begin{bmatrix} a + b & a - b \\ c + d & c - d \end{bmatrix} [a + bc + da - bc - d]$$

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$$\begin{bmatrix} a + b & a - b \\ c + d & c - d \end{bmatrix} [a + bc + d - a - b - c - d]$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} a + b & a - b \\ c + d & c - d \end{bmatrix} [a + bc + da - bc - d]$$

1 point

What is the matrix representation of T with respect to γ for both the domain and the co-domain?

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$$\begin{bmatrix} a + b + c + d & a - b + c - d \\ a - b - c - d & a - b - c + d \end{bmatrix} [a + b + c + da + b - c - da - b + c - da - b - c + d]$$

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$$\frac{a + b + c + da - b + c - d}{2} \frac{a - b - c - da - b - c + d}{2} [2a + b + c + d \ 2a + b - c - d \ 2a - b + c - d \ 2a - b - c + d]$$

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$$\frac{a + b + c + da + b - c - d}{2} \frac{a - b - c - da - b - c + d}{2} [2a + b + c + d \ 2a - b + c - d \ 2a + b - c - d \ 2a - b - c + d]$$

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$$\begin{bmatrix} a + b + c + d & a - b + c - d \\ a - b - c - d & a - b - c + d \end{bmatrix} [a + b + c + da - b + c - da + b - c - da - b - c + d]$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{a + b + c + da - b + c - d}{2} \frac{a - b - c - da - b - c + d}{2} [2a + b + c + d \ 2a + b - c - d \ 2a - b + c - d \ 2a - b - c + d]$$

1 point

Which of the following options are correct?

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If $T: R^2 \rightarrow R^2$ is a linear transformation which is both one-one and onto then $T^{-1}T = I$

is also a linear transformation.



If $T: R^n \rightarrow R^m$ $T: R_n \rightarrow R_m$ then the matrix corresponding to T^T , with respect to the standard bases for R^n R_n and R^m R_m is of order $n \times m$.



If $T: R^n \rightarrow R^m$ $T: R_n \rightarrow R_m$ then the matrix corresponding to T^T , with respect to the standard bases for R^n R_n and R^m R_m is of order $m \times n$.



Consider the vector space $V = \left\{ \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid a, b, c \in R, \right\}$ $V = \{[a0bc] \mid a, b, c \in R, \}$, that is, the set of all 2×2 upper triangular matrices with usual matrix addition and scalar multiplication. Then there exists an isomorphism from R^3 to V .

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $T: R^2 \rightarrow R^2$ $T: R_2 \rightarrow R_2$ is a linear transformation which is both one-one and onto then T^{-1} is also a linear transformation.

If $T: R^n \rightarrow R^m$ $T: R_n \rightarrow R_m$ then the matrix corresponding to T^T , with respect to the standard bases for R^n R_n and R^m R_m is of order $m \times n$.

Consider the vector space $V = \left\{ \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid a, b, c \in R, \right\}$ $V = \{[a0bc] \mid a, b, c \in R, \}$, that is, the set of all 2×2 upper triangular matrices with usual matrix addition and scalar multiplication. Then there exists an isomorphism from R^3 to V .

Check Answers

Your score is: 0/10